Hengyu Jin

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EDUCATION

Tongji University (TJU)
Major: Bachelor of Engineering, Software Engineering | **GPA:** 4.60/5.00 (91.01/100) Sep 2023 – Present

Research Interests: Reasoning and Reliability in Foundation Models; AI Agents

PUBLICATIONS

- **Jin, H.**, Yang, S., & Wang, D. **Final Checkpoints Are Not Enough: Analyzing Latent Reasoning Faithfulness Along Training Trajectories.** Under review at NAACL. arXiv:2607.06648.
- Ye, W., Yuan, X., Bin, Y., **Jin, H.**, Peng, L., Zeng, P., & Shen, H. T. **RISER: Orchestrating Latent Reasoning Skills for Adaptive Activation Steering.** Accepted to ACL Findings 2026. ACL.findings 2026.
- Chen, T., Chen, X., Shen, Y., Zhang, L., **Jin, H.**, & Cheng, C. **CORA: A Cognitive Reframing Dialogue Agent Powered by Large Language Models.** Accepted for publication in *IEEE Transactions on Affective Computing* (IEEE TAC). DOI: 10.1109/TAFFC.2026.3709312.
- Du, J., Zhang, Y., **Jin, H.**, Zou, X., Zhu, J., & Fu, X. **Kairotask: Probing the Bridge Between Vague Intents and Spatiotemporal Contexts.** Accepted to ACM CHI 2026. DOI: 10.1145/3772363.3798333.

RESEARCH EXPERIENCE

PRADA Lab, KAUST Jul 2026 – Present
Visiting Student | Advisor: **Prof. Di Wang** *Saudi Arabia*

Mechanistic Analysis of Latent Reasoning Faithfulness

- Led a first-author investigation of how latent reasoning faithfulness evolves throughout training, showing why final-checkpoint performance alone can mask changes in a model's internal reasoning process.
- Built a ProsQA evaluation pipeline combining BFS-verified counterfactual edits with latent-state interventions to test whether latent reasoning states causally reflect changes in problem structure across checkpoints.
- Compared latent and explicit reasoning approaches across diverse tasks, identifying distinct training dynamics that inform more reliable evaluation of reasoning systems.

Multi-User Agent Environment

- Developing a verifiable interactive environment that generates scenario-grounded multi-user rollouts for training and evaluating multi principal agent systems, addressing the scarcity and high cost of collecting real-world multi-user interaction data.

Aether AI Jun 2026 – Present
Research Intern | Advisor: **Prof. Biwei Huang, UC San Diego** *Remote*

Verifier for Robotics Agent System

- Developing an execution-time verifier for a robotics agent system to detect failures during VLA execution and trigger early termination, replanning, or tool-assisted recovery.
- Built a natural failure-data generation pipeline in Genie Sim by identifying atomic subtasks from simulator states, checkpointing key execution states, and resuming rollouts with weaker VLA policies to produce realistic failure trajectories.
- Standardizing evaluation across existing verifier and reward-model benchmarks to support joint training on internally collected and public failure data.

Tongji University Oct 2024 – Dec 2025
Research Intern | Advisor: **Prof. Ying Shen** *Shanghai*

Hidden State Steering to Enhance LLM Latent Reasoning Skills

- Co-developed a plug-and-play LLM activation steering framework that enhances reasoning capabilities, achieving 3.4–6.5% average accuracy improvement, with 2–3x token efficiency gains over Chain-of-Thought methods. by dynamically injecting cognitive primitive vectors into intermediate layers of a frozen model.
- Contributed to building a cognitive primitive construction pipeline based on Contrastive Activation Addition (CAA) + LLM-as-Judge filtering + KMeans clustering, distilling six categories of reusable reasoning vectors from Transformer hidden states.

CORA: CBT-Guided Post-Training for a Multi-Turn Dialogue Agent

- Fine-tuned Qwen2.5-7B-Instruct using LLaMA-Factory to adapt the model for multi-turn, CBT-guided cognitive reframing dialogues, contributing to the iterative development of the CORA dialogue agent.
- Designed and implemented a Streamlit-based evaluation platform for blinded comparisons across model variants and configurable inference settings, supporting experimental validation and dialogue-data collection.

Tsinghua University Jun 2025 – Oct 2025
Research Intern | Advisor: **Prof. Xinyi Fu** *Remote*

Human-Centered Design for Context-Aware Schedule Management

- To address the issue of disconnection between ambiguous intentions and execution scenarios, developed a backend reasoning architecture based on the large language model (Qwen-Plus) to achieve precise conversion from natural language to structured temporal metadata.
- Led the development of a complete mobile application by integrating iOS Core Location with geofencing technology, achieving location-based context-triggered notifications and reminders, and supporting an eight-day field deployment study.

PROJECTS

Co-founder & Full Stack Development Lead | Follo AI: The Next-Generation Context-Aware Intelligent Schedule Assistant Nov 2025 – Mar 2026

- Aimed to break through the reactive recording mode of traditional tools and build a dual-modal AI system that can simultaneously process user instructions and actively sense environmental changes.
- Designed the core "response–perception–memory" three-layer architecture: developed an intelligent router engine based on LLM to handle explicit instructions; wrote a multimodal perception model for the iOS end to collect 21-dimensional sensor data in real time to support situational decision-making; designed a structured memory mechanism to continuously optimize the user profile.
- Successfully developed and delivered the high-fidelity minimum viable product, verifying the efficiency of the "linear pipeline + edge perception" path.
- Advanced to the financing phase and established the architectural groundwork for integration with wearable devices.

Team Lead & Core Developer | "Scrollscape" Vision Pro AR Application Jun 2025 – Sep 2025

- Spearheaded the development of an immersive Chinese garden education application for Apple Vision Pro, converting abstract aesthetic concepts into interactive spatial-computing experiences.
- Overcame the obstacles posed by high-precision gesture-interaction conflicts and key rendering technologies such as the "virtual–real portal" on the Vision Pro platform, and implemented an automated aesthetic evaluation system.
- Won the First Prize in the East China Region of the China Collegiate Computing Contest–Mobile App Innovation Division.

AWARDS AND HONORS

First Prize (East China Region), China Collegiate Computing Contest–Mobile App Innovation Division (2025)
First-grade Ethnic Class Special Scholarship, TJU (2023–2024, 2024–2025)

SKILLS

Relevant Coursework: Machine Learning, Computer Vision, Autonomous Speech Recognition, Computer Architecture, Computer Networks, Data Structures, Discrete Mathematics, Linear Algebra, Algorithm Design and Analysis
Languages: Chinese (Native); English (TOEFL: 4.5/6)
Programming Languages: Python / C++ / Swift / SwiftUI / React / FastAPI